

Taking Turns

Natural Language Inference on Interview Data

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1 NLI and interview transcripts

target: 50–70 words

Entailment is a relation between propositions. NLI scores most reliably where a claim is asserted or stated rather than conveyed, and when excerpts are short enough that claims occupy most of them. Interview transcripts are unexplicit, long, and multi-thematic — common properties of interview speech, but a combination that is difficult for NLI to score.

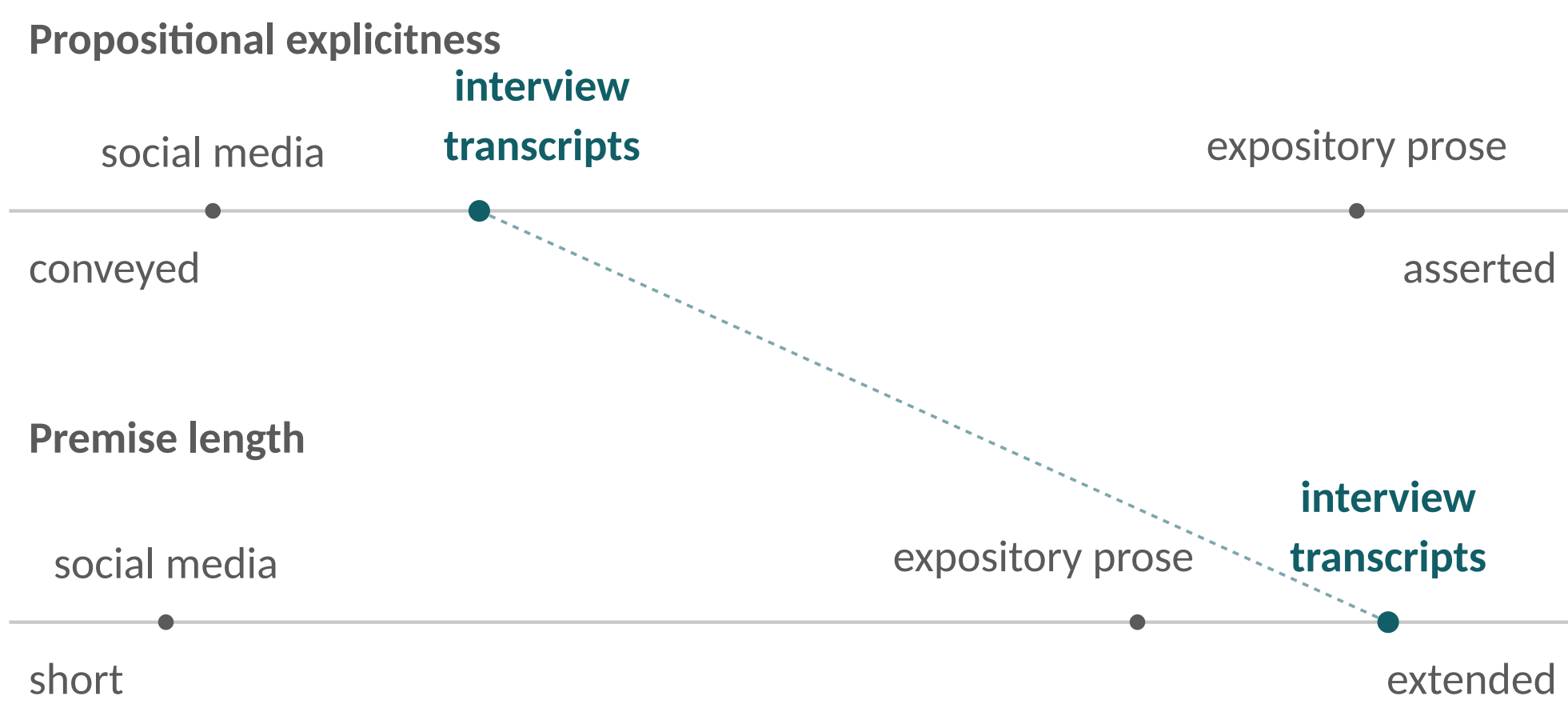


Figure 1. Transcripts sit with social media on explicitness and with expository prose on length.

2 Data and pipeline

Corpus. 44 in-depth interviews with LGBTQ activists; N = 1,964 individual responses (premises).

Hypotheses. 13 short declarative statements about how speakers evaluate movement outcomes.

Model. ModernCE-large-nli: cross-encoder, ~395M parameters, 8,192-token context window.

Inference. Every premise scored against every hypothesis for a total of 25,532 pairs.

Independence. Hypotheses are scored independently so that one premise can register several claims. Thresholds are set per-hypothesis using close reading.

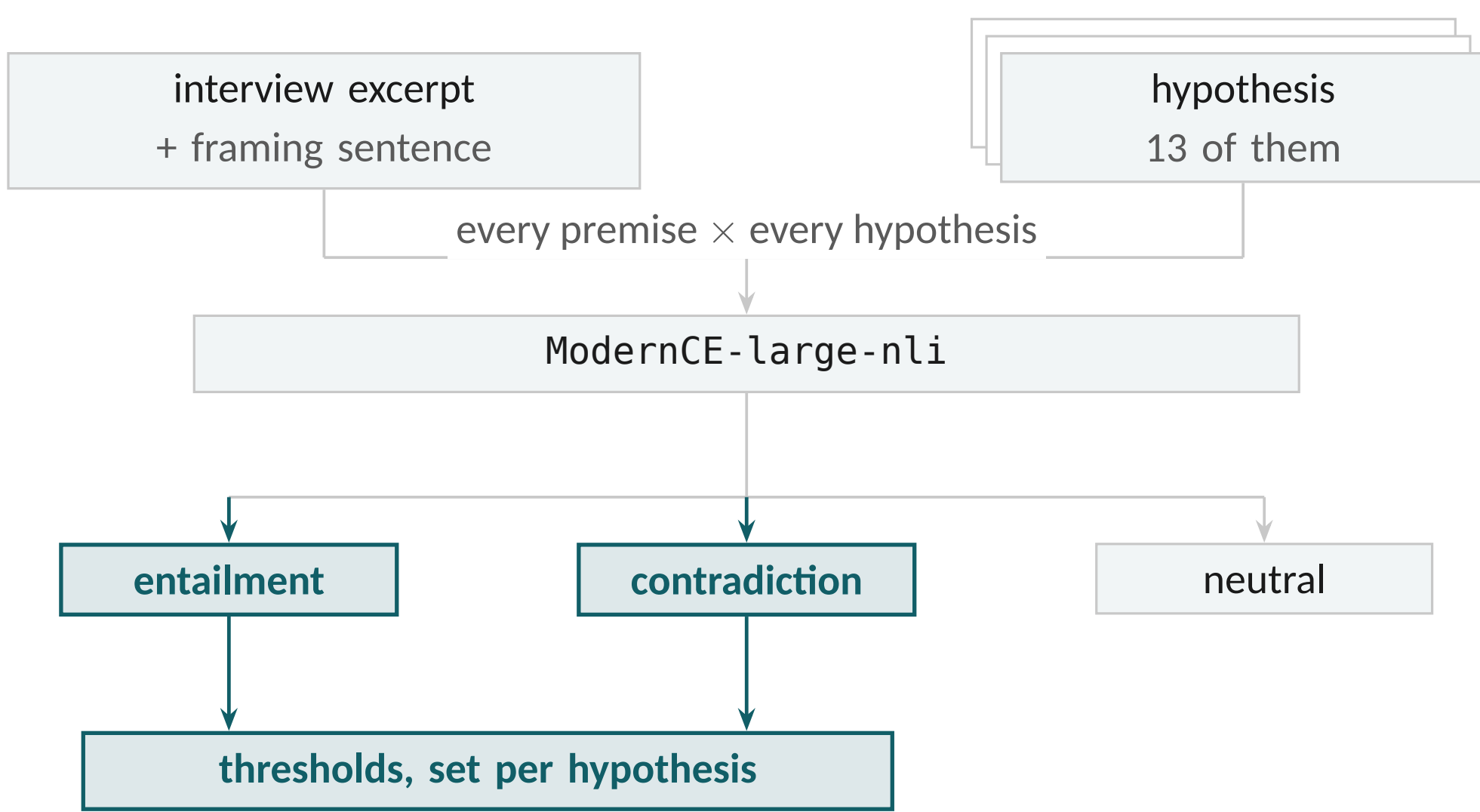


Figure 2. Entailment and contradiction are thresholded separately for every hypothesis.

3 Hypotheses

Framing sentence prepended to every premise:
“In this excerpt, an LGBTQ activist discusses their experiences and views.”

Framing sentence is minimal, neutral, and provides enough context for the model to resolve contextual cues without biasing it toward any hypothesis.

positive_change
“The speaker describes marriage equality as a positive change.”

prioritize_beyond
“The speaker argues that the LGBTQ movement needs to prioritize issues beyond marriage equality.”

joy_celebration
“The speaker expresses joy or celebration about the achievement of marriage equality.”

4 Multi-thematic excerpts

“Marriage Equality, it was great. It was a celebration. It was a huge victory. But at the same time, it only covers, like, the aspect of marriage. Right? We still need to keep folks protected from losing their jobs. Right? Like, there’s so much more that we need to do.”

— Amanda

Hypotheses clearing threshold on this one excerpt:

- prioritize_beyond 0.92
- positive_change 0.91
- joy_celebration 0.82

Amanda makes three claims in four sentences: marriage equality was a victory, it was a cause for celebration, and it was not enough. Single-label classification would have to pick one, and any choice would misrepresent the passage.

A protocol for NLI on interview data

1. **Prepend a minimal framing sentence.** Enough context to resolve indirect reference and neutral enough not to tilt any hypothesis.
2. **Segment with dilution in mind.** Long premises buy context up to a point, then cost signal.
3. **Calibrate thresholds per hypothesis** by close reading of interview excerpts across the entailment distribution. Do not rely on a global cut-off.
4. **Read the contradiction side as well.** Contradiction surfaces real signal at the opposite pole of the hypothesis, and has significant analytical utility.

Naturalistic prose (interviews, life histories, testimonials, field notes) is foundational to interpretive social science. NLI provides a promising method for systematic, reproducible classification of naturalistic prose that is sensitive to its multi-thematic nature.

5 Hypotheses travel in families

Excerpts rarely register a single claim. Because each hypothesis is scored independently, what co-occurs is a property of the speech rather than of a coding scheme that forced a choice.

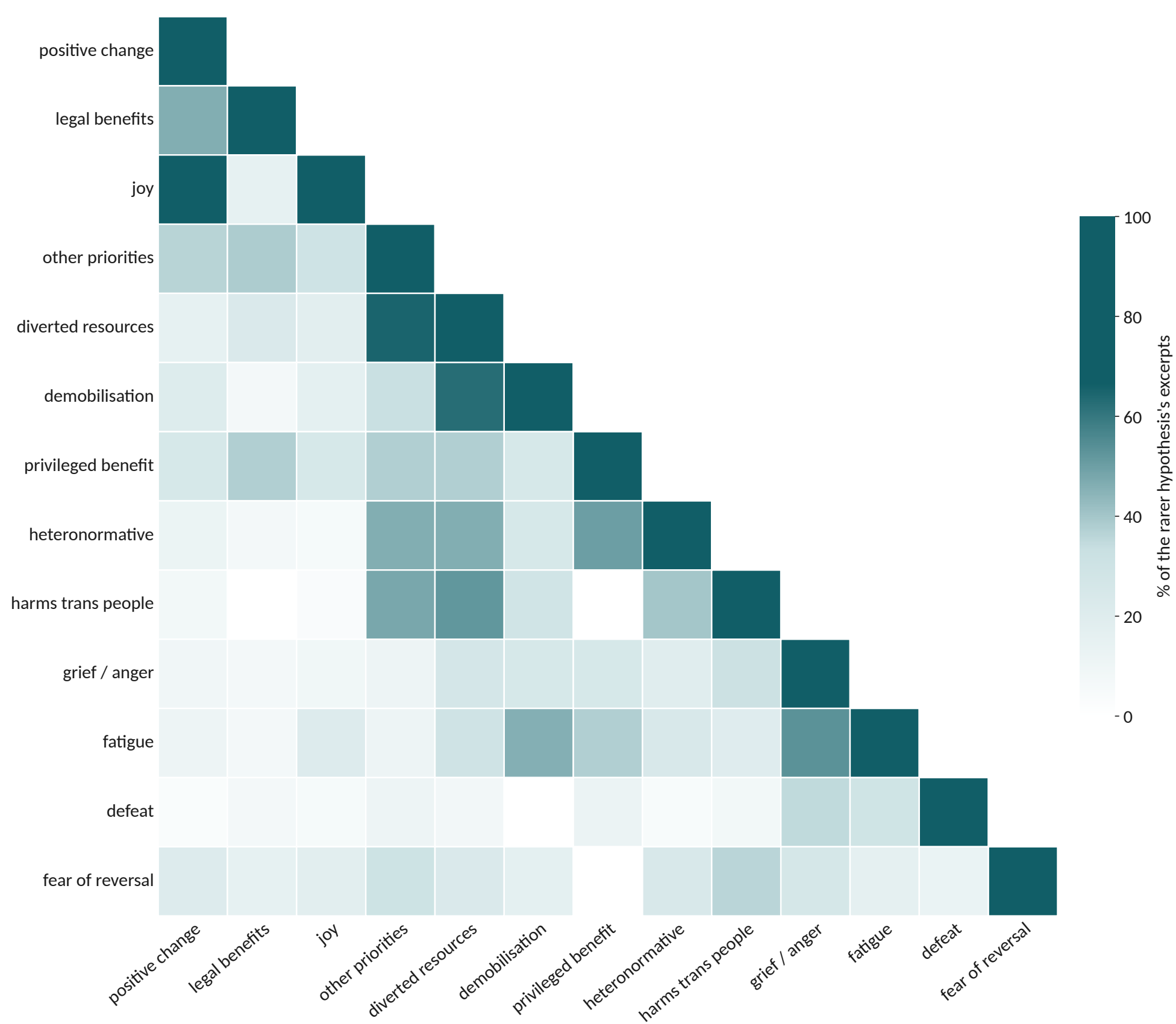


Figure 3. Co-occurrence is patterned. Celebratory claims recur together, critical assessments recur together, and grief, fatigue and defeat form a third family.

Each cell reports the share of the rarer hypothesis’s above-threshold excerpts that also clear threshold on the other, which controls for differing base rates.

- Three families of hypotheses are visible on the diagonal: supportive, critical, negative-affective.
- Because hypotheses are scored independently, this structure emerges from the data rather than being imposed by a coding scheme.
- Roughly half the sample produces excerpts in both the supportive and critical families. The distinction is *between discursive modes*, not between types of people.

6 Segmentation trade-off

The model scores each premise holistically. A relevant claim occupying a small fraction of a long, multi-theme response is diluted. This sets premise length against signal.

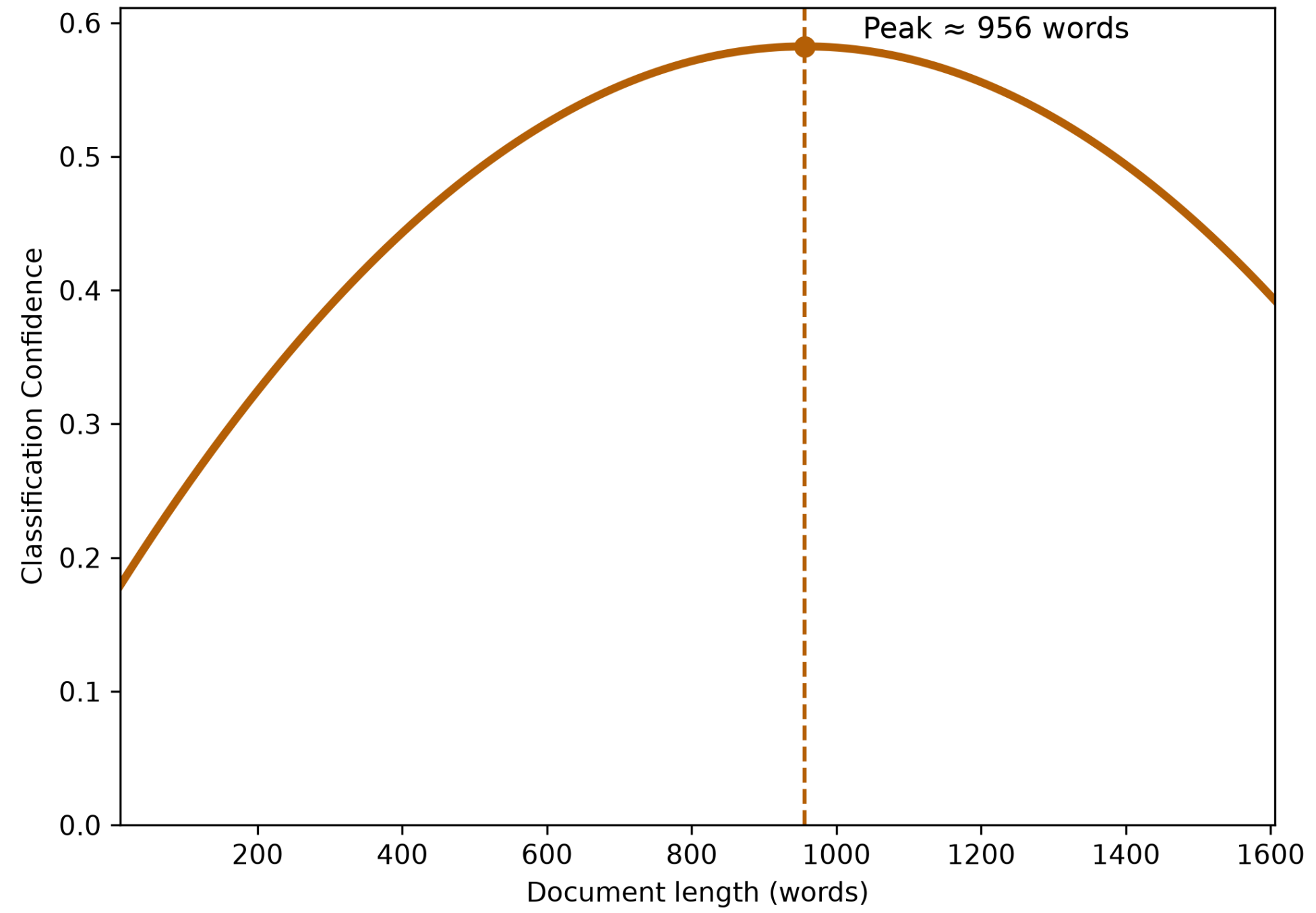


Figure 4. Confidence rises with context, then falls as the relevant claim is diluted. Estimated turning point at 956 words (N = 1,964).

The turning point is corpus-specific, but transcript segmentation should be designed with this trade-off in mind.

7 Recurring failure modes

- Affect without object**
The model detects that emotion is present far more reliably than what the emotion is about. Affective hypotheses fire broadly, while propositional ones fire precisely.
e.g. conflating joy over any victory with joy over marriage equality.
- Speaker attribution**
False positives wherever a participant narrates someone else’s experience or takes on a persona. The model scores the claim, not who is making it.
e.g. narrating opposition claims.

Register and vocabulary mismatch
Model performs better when hypothesis and premise employ similar registers and vocabularies, and underperforms when the model needs to bridge register/vocabulary gap.
Entailments run lower for participants using unique vocabulary.

Consequence: A global per-hypothesis threshold may not be defensible when corpus is built from diverse population with unique linguistic features.